

NILPOTENT GROUPS

- *(N1) (A.Miasnikov) Let G be a free nilpotent group of finite rank. Suppose an element $g \in G$ is fixed by every automorphism of G . Is it true that $g = 1$? [Background](#)
- *(N2) Let G be a finitely generated nilpotent group. Is the Dehn function of G equivalent to a polynomial? [Background](#)
- (N3) (B.I.Plotkin) Is it true that every locally nilpotent group is a homomorphic image of a torsion-free locally nilpotent group?
- (N4) (G.Baumslag) Let G be a finitely generated torsion-free nilpotent group. Is it true that there are only finitely many non-isomorphic groups in the sequence $\text{Aut}(G), \text{Aut}(\text{Aut}(G)), \dots$? [Background](#)
- (N5) (G.Baumslag) Is the property of being directly indecomposable decidable for finitely generated nilpotent groups?
- (N6) (A.Miasnikov) Describe all finitely generated nilpotent groups of class 2 which have genus 1. (We say that a group G has genus 1, if every group with the same set of finite homomorphic images as G , is isomorphic to G).
- (N7) Is every group with an Engel identity $[x, y, \dots, y] = 1$, locally nilpotent? [Background](#)
- (N8) (A.Miasnikov) * (a) Is it true that equations of the form $[x, y] = g$ ($g \in G$) are decidable in every finitely generated 2-nilpotent group G ?
(b) Are equations of the form $[x, y] = g$ ($g \in G$) decidable in every finitely generated free nilpotent group? [Background](#)
- (N9) (A.Miasnikov) Let G be a group. The *retract problem* in G is the following: given a finitely generated subgroup H of G , decide if H is a retract of G or not.
(a) Is the retract problem decidable in every finitely generated nilpotent group G ?
*(b) Is the retract problem decidable in finitely generated free nilpotent groups? [Background](#)